1928
K. N. Toosi University of Technology

Recursion – Homework

Advanced Progamming

by

F.Ahmadi

Q1: Recursive String Compression

K. N. Toosi University of Technology

- Write a recursive function to compress a string by counting consecutive characters.

```
def compress_string_recur(s: str) -> str:

…

Examples :


compress_string_recur("aaabbc") → "a3b2c1"

compress_string_recur("abcd") → "a1b1c1d1"
```

Q2 : Deep Sum of Nested Lists

- Write a recursive function to sum of all integers inside a nested list of arbitrary depth.

```
def deep_sum(data):
…
```

```
Examples:
deep_sum([1, [2, [3, 4], 5], 6]) → 21
deep_sum([[[1]], 2, [[3, [4]]]]) → 10
```

Q3: Maximum Depth of Nested Lists

1928
K. N. Toosi University
of Technology

- Compute **maximum depth** of a nested list:

```
def deep_max_depth(data):




Example:

deep_max_depth([1, [2, [3, [4]]]]) → 4
```

Debugging Recursive Functions (Fix the bugs)

1928
K. N. Toosi University of Technology

- **find and fix the bugs in the following recursive function:**

```python
def flatten(lst):
    if not lst:
        return []
    if isinstance(lst[0], list):
        return flatten(lst[0])
    else:
        return [lst[0]] + flatten(lst[1:])
```